

Prior Work Assessment & What We Can Reuse

Project: मोडी ते मराठी (modi-to-Marathi) — Modi script → Marathi (Devanagari)
Repository: <https://github.com/lgtkgvtv/ai4-transliteration-modi-to-marathi> **From:** Sachin, Project Maintainer **To:** Project contributors and faculty advisors **Date:** [Date]
Re: What already exists publicly, how good it is, and how its quality was verified

A note before you read

Team — before we ask anyone to contribute images or spend expert hours, I wanted us to start from a shared understanding of what's already been done and published for exactly our task: transliterating Modi script into Marathi (Devanagari). Below is my current reading of the most relevant prior project. **Several of you have researched this topic directly, so please treat this as a draft for you to correct** — where I've got something wrong or oversimplified, I'd rather hear it now than after we've built on a bad assumption.

This document answers three questions together: 1. What can we reuse instead of rebuilding? 2. How good is it, really? 3. How did they prove it was good — and how would we check that ourselves?

1. The reference project

The closest published work to ours is from **IIT Roorkee**: "Historic Scripts to Modern Vision: A Novel Dataset and a VLM Framework for Transliteration of Modi Script to Devanagari." It targets the same source→target we do.

Detail	Value
Authors	H. Kausadikar, T. Kale (co-first), O. Susladkar, S. Mittal
Preprint	arXiv:2503.13060
Peer-reviewed	ICDAR 2025, Springer LNCS, DOI 10.1007/978-3-032-04630-7_3

Detail	Value
Public org	historyHulk on Hugging Face

It's the first published attempt to transliterate **whole handwritten Modi documents** (not just isolated characters) directly into Devanagari — which is precisely our goal.

2. What they open-sourced, and how we'd access it

Artifact	What it is	Access
historyHulk/MoDeTrans	~2,043 real Modi manuscript images + verified Devanagari	Public
historyHulk/SynthMoDe	Synthetic Modi images rendered from the same Devanagari text	Public
historyHulk/ModiTrans-12B-Gemma-Teacher	A model adapter (built on Google's Gemma-3-12B)	Gated — free account + accept terms

Their dataset composition tells us which eras they prioritized — useful as a reference point for our own collection:

Era	Images	Share
Shivakalin	891	43.6%
Peshwekalin	944	46.2%
Anglakalin	208	10.2%

They split the data 80:10:10 (train/test/validation). Notably, they did **not** include the older Adyakalin and Yadavkalin eras, because well-preserved documents from those periods are scarce — a constraint we'll likely share.

3. An important practical caveat (please sanity-check me on this)

There's a gap between what the press coverage implies and what's actually downloadable, and we should be clear-eyed about it:

- The paper's headline model is a **small, efficient 429M-parameter model** ("MoScNet-XL") trained by distilling a very large teacher.
- But the model actually **published** is the **large 12B teacher**, not that small student — and it's built on a different base model than the paper's headline configuration.
- As far as I can find, **the lightweight model the announcements emphasized is not yet downloadable.**

Why this matters for us: if we plan to "just reuse their model," we'd actually be running a heavy 12B model, and we shouldn't assume it reaches the accuracy numbers quoted in the paper. **Advisors — if any of you know whether the smaller model has since been released, please flag it.**

4. How they verified efficacy and correctness (the part I most want your eyes on)

Their credibility rests on how they measured success, and we should hold ourselves to the same or higher standard.

Dataset integrity measures they used: - Transliterations were done by **qualified Modi-script experts**, with written guidelines to keep them objective. - A **cross-verification** pass by experts to catch and fix transliteration errors. - **10 annotators manually checked** the train/test/validation splits to ensure **no overlap** (no data leakage), on top of automated splitting. - They used **only genuine historical documents** from credible archives (e.g. the Bharat Itihas Sanshodhak Mandal in Pune, archives departments) — not invented text — for the real dataset.

How they measured model quality:

Method	What it checks
BLEU score	

Method	What it checks
	Overlap between the model's output text and the expert reference
Character-level accuracy (on standard OCR benchmarks)	Reading accuracy at the character level
Baselines	Compared against many prior methods (older neural nets through large language models)
Ablation studies	Removed parts of their own system to prove each part actually helps
Zero-shot generalization test	Trained on real data, tested on synthetic data, to show it didn't just memorize

The takeaway: they didn't report a single number — they **triangulated** with multiple metrics, comparisons, and stress tests. That's the bar for us.

5. How we would independently verify their claims (proposed checklist)

If we build on their work, I propose we confirm it ourselves rather than take it on faith:

1. Re-run their released model on **their own published test split** and compute the scores ourselves.
2. Add a **character-level error rate** (how many characters are wrong), which is arguably more honest for transliteration than BLEU alone.
3. Test it on a **small sample of our Modi documents** that it has never seen, to see how it holds up outside their data.
4. Independently check for **data leakage** between splits before trusting any score.
5. Have **one of our experts spot-check** outputs for the error types that matter to us — proper nouns, place names, numbers and dates in land records.

Advisors: anything you'd add from your own experience reviewing or doing this kind of work?

6. The Modi-specific hard cases we already know about

From their published error analysis, the model reliably struggles with visually similar Modi characters and diacritics. We should expect — and specifically test for — these:

- भ / म (bha / ma)
- क / फ (ka / pha)
- ट / ठ / ढ (ṭa / ṭha / ḍha)
- न / ण (na / ṇa)
- Vowel-sign slips such as के / कि (ke / ki)
- Dropped **anusvāra** (the dot above a character), which changes pronunciation

There are also structural challenges intrinsic to Modi: a continuous shirorekha (top line), **no spaces between words**, and a cursive, angular style. These make segmentation genuinely hard and are worth raising with our experts early.

7. Evidence that this class of approach genuinely works

So we're not relying on a single project's claims, here are three independent, peer-reviewed efforts in the same family — useful both as proof-of-concept and as models of rigorous evaluation:

Project	Task	Published in	Result & lesson
Ithaca (DeepMind + Venice/Oxford/Athens)	Restoring ancient Greek inscriptions	Nature 2022	Historians' accuracy rose from 25% → 72% with the AI. Lesson: the AI is most valuable assisting an expert, and they measured the combined human+AI score.
Akkadian → English (Guthertz et al.)	Cuneiform translation	PNAS Nexus 2023	Worked well on short text, openly reported where it fails. Lesson: publish your failure modes, design a correction loop.

Project	Task	Published in	Result & lesson
IndicXlit / Aksharantar (AI4Bharat, IIT Madras)	Transliteration across 21 Indic languages	EMNLP Findings 2023	A tiny (~11M) model on a huge open dataset (26M pairs) , now deployed nationally. Lesson: for transliteration, the dataset is the moat , not model size.

The recurring theme — and the most important point in this document — is that **the dataset and the human-in-the-loop process, not the model, are what make or break these projects**. That's why our first real work is dataset preparation (see `docs/03_Dataset-Preparation-Plan.md`).

8. What I need from the faculty advisors specifically

When you have time, I'd value your input on: - Anything factually off in sections 3–6 above. - Prior datasets, tools, or your own past work we should reuse or learn from. - Whether the verification checklist in section 5 is sufficient. - Pitfalls you hit in earlier research that we should design around from day one.

Thank you — I'd rather invest this alignment time now than discover gaps later.

-sachin